Rishi Dhar

Doctoral researcher, Institute of Aerospace Thermodynamics (ITLR), University of Stuttgart
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Profile

PhD researcher running direct numerical simulations of thixotropic jet atomization on GPU supercomputers at HLRS, within a DFG project on primary and secondary breakup of shear-thinning and thixotropic jets. Background in multiphase CFD, spray and combustion modelling, and HPC; currently moving towards GPU engineering (modern C++, CUDA, open-source HPC).

Experience

Doctoral Researcher Feb 2024 – present
ITLR, University of Stuttgart

- DNS of thixotropic liquid jet atomization with the in-house VOF solver FS3D; supervised by Prof. Bernhard Weigand and Dr. Kathrin Schulte.
- GPU porting of FS3D with OpenMP target offloading on AMD MI300A (HLRS Hunter); post-processing pipelines for multi-terabyte HDF5 output on HLRS Hawk and Hunter.
- Validated turbulent inflow-profile solvers for shear-thinning and Moore-model fluids.

Intern, then Master's Thesis Mar 2023 – Nov 2023
Siemens, Munich

- Nano-scaled model of cathode slurry components (PVDF, carbon black, NMC) with validation against experiments.
- Viscosity-versus-shear analysis of non-Newtonian fluids using CFD coupled with the discrete element method (DEM); battery slurry mixing models in STAR-CCM+, NX and OpenFOAM; thermal-runaway simulation of Li-ion NMC cathodes.

Student Researcher 2022 – 2023
Institute for Combustion Technology (ITV), RWTH Aachen

- Fortran spray simulations on the RWTH cluster: reduced-order models for multi-fuel, multiphase injection, validated with ECN and in-house experimental data.

Student Researcher Dec 2022 – Apr 2023
Institute of Jet Propulsion and Turbomachinery (IST), RWTH Aachen

- LES of compressor blade cascades with periodic boundary conditions (TRACE) in coordination with MTU; post-processing in Tecplot/PyTecplot; data filtering and validation in Python.

Management Trainee to Assistant Manager (Design and Thermal Engineer) 2017 – 2021
Bharat Petroleum (BPCL), Surat / Vadodara

- Integrated existing automation and safety PLC systems; managed a terminal expansion project (600 Cr INR capex).
- Thermal validation simulations for the flameproof rating of rubber safety nets on hydrocarbon sensors and blow-valves; DFM/DFA of solenoid valves and piezo-electric pipeline sensors.

Summer Intern May – Jul 2016
Hero MotoCorp, Gurgaon

- Designed and built an engine test bench, enabling 100% quality certification of engine exports.

Education

M.Sc. Computer Aided Conception in Mechanical Engineering 2021 – 2023
RWTH Aachen University

- Focus on CFD, multiphase flow and turbulence modelling. Grade: 2.0.

B.Tech. Mechanical Engineering 2013 – 2017
MANIT Bhopal

- Top 10% of class. Grade: 1.9. Thesis advised by Prof. Rajesh Gupta.

Publications and Talks

The Role of Interfacial Tension in Direct Numerical Simulations of Drop-Film Interaction for Immiscible Fluids 2026
R. Dhar, D. Gösele, P. Saumet, B. Weigand, K. Schulte. arXiv:2605.30082

Spatial regime structure in direct numerical simulation of thixotropic liquid jets in preparation
R. Dhar, K. Schulte, B. Weigand. Journal paper

Investigation of Droplet-Pool Phenomena and Droplet-Film Interaction for Immiscible Fluids 2024
R. Dhar, K. Schulte. DIPSI 2024

CFD DEM Coupling for Battery Slurry 2024
R. Dhar. ITV, RWTH Aachen

A Reduced-Order Model for Multiphase Simulation of Transient Inert Multicomponent Sprays in the Context of Compression Ignition Engines 2023
A. Saha, M.S. Deshmukh, A.Y., R. Dhar, et al. IACM CFC 2023, talk

A Study on the Design Procedure: Geometry, Ability to Endure and Perform and Sustainance 2016
R. Dhar, A. Gupta, M. Shukla, A. Sharma, A. Porwal, R. Biswal. Imperial Int. J. of Eco-friendly Technologies 1(1), 8

Design optimization of a nose profile for a canard configuration missile
K. Akojwar, R. Dhar

Selected Projects

GPU offloading of FS3D 2026
Fortran, OpenMP target, MI300A

- Porting the VOF solver to AMD MI300A APUs; benchmarking interface advection against PARIS.

Jet DNS post-processing pipeline 2026
Python, h5py, ParaView

- Grid-convergence (GCI), Kolmogorov-resolution and statistical-convergence diagnostics; phase-conditioned ParaView batch visualisation.

Rust image transfer for helm-dtn 2026
Rust, Go, Kubernetes

- AAP2 sender/receiver over µD3TN for moving container images across delay-tolerant links.

Design Lead, Electric Solar Vehicle Championship 2015 – 2016
MANIT Bhopal

- Led a 5-member team; designed a low-drag chassis and roll cage. Won the Best Design Award among 150+ universities.

Skills

CFD and numerics	DNS, volume-of-fluid multiphase flow, non-Newtonian and thixotropic rheology, LES/RANS, spray and combustion solvers, multigrid, Lattice Boltzmann
HPC and GPU	Fortran, MPI, OpenMP target offloading, HLRS Hawk/Hunter (PBS), HDF5/XDMF; learning: modern C++, CUDA, Kokkos, AMReX
Tools	Python (NumPy, h5py, SciPy), ParaView, Tecplot, STAR-CCM+, OpenFOAM, Rust, Go, Kubernetes, LaTeX/TikZ, Git, Linux
Languages	English, Hindi, German